Reimplementation and Improvement of ViHateT5 for Vietnamese Hate Speech Detection

DS200.Q21 - Big Data Analysis - Group 02
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Outline

1. Introduction
2. Datasets
3. Methodology
4. Experiments
5. Results
6. Limitations & Future Work
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Problem Statement

Hate Speech on Social Media

- Rapid growth of Vietnamese social media content
- Hate speech, offensive language, and toxic comments spread quickly
- Manual moderation is **infeasible** at scale
- Automated detection systems are critical

Challenges

- Multiple hate speech **tasks** (classification, span detection)
- Limited labeled data for Vietnamese
- Class imbalance in existing datasets

Existing Datasets

- **ViHSD**: Hate speech (3 classes)
- **ViCTSD**: Toxic speech (binary)
- **ViHOS**: Hate spans
- **VOZ-HSD**: 10M+ unlabeled forum posts

Goal

Build a **unified** model for Vietnamese hate speech detection across multiple tasks and datasets.

Paper Overview

Original Paper – ViHateT5 (ACL 2024 Findings)

Luan Thanh Nguyen, "*ViHateT5: Enhancing Hate Speech Detection in Vietnamese With a Unified Text-to-Text Transformer Model*," Findings of ACL 2024, pp. 5948–5961.

Key Contributions

1. Unified **text-to-text** formulation for 3 Vietnamese HSD tasks
2. **Continual pre-training** with Span Corruption on domain data
3. Multi-task Seq2Seq fine-tuning using **ViT5-base**

Our Contributions

1. Full **reimplementation** of ViHateT5
2. **Auto-labeling pipeline** with ViSoBERT (10M+ samples, 97.5% accuracy)
3. Extended **BERT baselines** (8 models)
4. **Data-centric improvement** beyond the original paper

Project Objectives

Course Requirements (DS200.Q21)

1. **Understand** the algorithms and AI models used for data analysis
2. **Design and reimplement** the paper's methods
3. **Evaluate** strengths and weaknesses of the approach
4. **Propose improved solutions** beyond the original paper

What We Reproduced

- T5 continual pre-training
- Multi-task Seq2Seq fine-tuning
- BERT-based classification baselines
- All result tables (Tables 3–6)

What We Improved

- Auto-labeling pipeline for VOZ-HSD
- 200K balanced pre-training corpus
- Extended comparison with 8 BERT models
- All models hosted on HuggingFace

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Datasets Overview

Dataset	Type	Labels	Description
ViHSD	Multi-class (3)	CLEAN / OFFENSIVE / HATE	Vietnamese hate speech detection
ViCTSD	Binary	NONE / TOXIC	Vietnamese toxic speech detection
ViHOS	Hate Spans	Character-level tags	Hate span detection
VOZ-HSD	Binary	Auto-labeled	10M+ VOZ forum comments

Table: Datasets used in this project

Data Sources

- ViHSD, ViCTSD, ViHOS: public benchmarks from HuggingFace
- VOZ-HSD: large-scale unlabeled data from VOZ forum
- All loaded automatically via `data_loader.py`

Key Challenge

- VOZ-HSD is **unlabeled** (10M+ samples)
- Manual annotation is infeasible
- Solution: **auto-labeling** with ViSoBERT
- 97.5% agreement with manual labels

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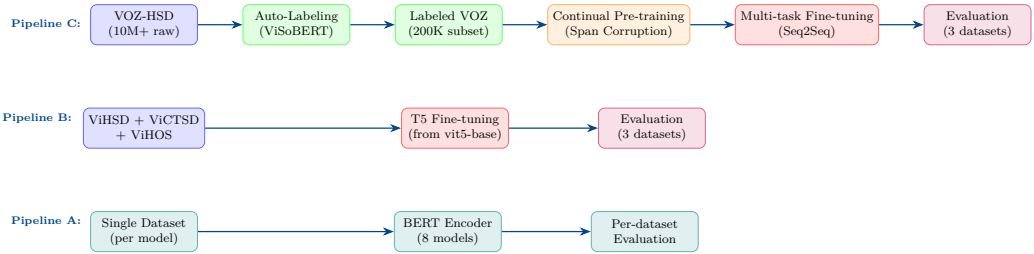
Overall Pipeline



- **Pipeline A**: BERT baselines (8 encoders, per-dataset)
- **Pipeline B**: T5 text-to-text (ViHateT5, multi-task)
- **Pipeline C**: Data-centric improvement (auto-label → pre-train → fine-tune)

Approach 1: BERT-based Classification (Baseline)

Architecture

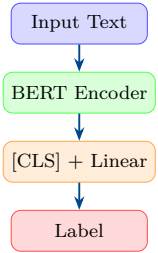
- Standard encoder-only transformers
- Pre-trained encoder + classification head
- Fine-tuned **independently** on each dataset

Models Evaluated

- **PhoBERT** (base, v2, large)
- **ViSoBERT** (best performer)
- **XLM-RoBERTa** (base, large)
- **BERT** (multilingual cased/uncased)
- **DistilBERT**, **viBERT**

Training Configuration

- Batch size: 16
- Learning rate: 2×10^{-5}
- Epochs: 10
- Early stopping: patience = 3
- Optimizer: AdamW
- Max length: 256 tokens



Approach 2: T5 Text-to-Text (ViHateT5)

Text-to-Text Formulation

All tasks are formulated as sequence-to-sequence:

- **ViHSD**: `hate-speech-detection: <text>` → `CLEAN` / `OFFENSIVE` / `HATE`
- **ViCTSD**: `toxic-speech-detection: <text>` → `NONE` / `TOXIC`
- **ViHOS**: `hate-spans-detection: <text>` → text with `[HATE]...[HATE]` tags

Advantage

A **single model** handles all three tasks via task-specific prefixes — no need for separate classifiers.

Training Configuration

- Base model: **VietAI/vit5-base**
- Batch size: 32
- Learning rate: 2×10^{-4}
- Epochs: 4
- Weight decay: 0.01
- Best model metric: Macro F1
- Mixed precision: bf16

Multi-task Training

All three datasets are **concatenated** and trained **jointly** in a single run. Evaluation is performed per-dataset.

Approach 3: Data-Centric Improvement

Key Innovation Beyond Original Paper

Auto-labeling + continual pre-training on large-scale domain data before fine-tuning.

Step 1: Auto-Labeling

- Fine-tuned **ViSoBERT** labels 10M+ VOZ comments
- 97.5% agreement with manual labels
- Batch inference on H200 GPU

Step 2: Pre-training

- **Span Corruption** objective on labeled VOZ data
- Two setups:
  - 100K hate-only
  - 200K balanced
- 10 epochs, lr $= 5 \times 10^{-3}$

Step 3: Fine-tuning

- Fine-tune from **domain checkpoint**
- Same multi-task Seq2Seq setup
- Improved contextual understanding

ViSoBERT labels VOZ → Span Corruption → Seq2Seq Fine-tune → Evaluation

Span Corruption Pre-training

How Span Corruption Works

1. Randomly select **15%** of tokens as noise
2. Replace consecutive corrupted tokens with **sentinel tokens** (`<extra_id_0>`, `<extra_id_1>`, ...)
3. Model learns to **predict the original spans**
4. Mean noise span length: 3.0 tokens

Example

Input: "Tôi <extra_id_0> Việt Nam <extra_id_1>"

Target: "<extra_id_0> yêu <extra_id_1> rất nhiều"

Pre-training Configuration

- Model: **VietAI/vit5-base**
- Noise density: 0.15
- Mean span length: 3.0
- Batch size: 128
- Learning rate: 5×10^{-3}
- Epochs: 10
- Warmup steps: 2000
- Optimizer: AdamW
- Gradient checkpointing: enabled
- Mixed precision: bf16

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Training Configurations Summary

Stage	Base Model	Batch Size	LR	Epochs	Optimizer
Pre-training T5	vit5-base	128	5×10^{-3}	10	AdamW
Fine-tuning T5	Pre-trained checkpoint	32	2×10^{-4}	4	AdamW
Training BERT	phobert / visobert	16	2×10^{-5}	10	AdamW
Auto-Labeling	visobert (fine-tuned)	128	—	—	—

Table: Training configurations for all stages

Hardware

- **NVIDIA H200** (141 GB) – Pre-training & auto-labeling
- **P100** – BERT fine-tuning
- GPU provided by FPT via voucher

Reproducibility

- All scripts in `scripts/` directory
- Configurable CLI arguments
- Random seed: 42
- Models & data on HuggingFace

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Auto-Labeling Performance

ViSoBERT Auto-Labeling

Metric	Result
Total samples labeled	10,747,733
Agreement w/ manual	**97.5%**
Accuracy	97.5%
Hardware	H200 GPU

Significance

- ViSoBERT achieves **97.5% accuracy** compared to original manual labels
- Enables large-scale pre-training without manual annotation
- Two pre-training subsets created:
  - **100K** hate-only samples
  - **200K** balanced samples
- Dataset published on HuggingFace

BERT Results – Per Dataset (Table 3)

ViHSD

Model	Acc	F1
ViSoBERT	.8842	**.6871**
PhoBERT v2	.8725	.6583
XLM-R	.8692	.6544
BERT (cased)	.8665	.6427
PhoBERT	.8632	.6360
DistilBERT	.8615	.6224
BERT (uncased)	.8561	.6161
viBERT	.8596	.6149

ViCTSD

Model	Acc	F1
PhoBERT	.9078	.7131
XLM-R	.9015	**.7153**
PhoBERT v2	.9023	.7139
ViSoBERT	.9035	.7045
DistilBERT	.8962	.6850
BERT (uncased)	.8993	.6796
BERT (cased)	.8983	.6710
viBERT	.8881	.6765

ViHOS

Model	Acc	F1
ViSoBERT	.9016	**.8578**
XLM-R	.8834	.8133
BERT (cased)	.8601	.7637
DistilBERT	.8585	.7615
BERT (uncased)	.8520	.7393
PhoBERT v2	.8492	.7351
viBERT	.8463	.7291
PhoBERT	.8465	.7281

ViSoBERT achieves the best average Macro F1 = **0.7498** across all three datasets.

BERT Average Macro F1 Comparison

Model	ViHSD F1	ViCTSD F1	ViHOS F1	Avg F1
ViSoBERT	0.6871	0.7045	0.8578	**0.7498**
XLM-RoBERTa	0.6544	0.7153	0.8133	0.7277
PhoBERT v2	0.6583	0.7139	0.7351	0.7024
BERT (cased)	0.6427	0.6710	0.7637	0.6925
PhoBERT	0.6360	0.7131	0.7281	0.6924
DistilBERT	0.6224	0.6850	0.7615	0.6896
BERT (uncased)	0.6161	0.6796	0.7393	0.6783
viBERT	0.6149	0.6765	0.7291	0.6735
Overall Average	0.6412	0.6949	0.7660	0.7007

Table: Average Macro F1 across 3 datasets (Table 3)

T5 Models – Detailed Results (Table 4)

Model	Dataset	Accuracy	F1 Weighted	F1 Macro
ViT5 (Base)	ViHSD	0.8777	0.8787	0.6625
ViT5 (Base)	ViCTSD	0.9080	0.9178	0.7163
ViT5 (Base)	ViHOS	0.9075	0.9000	0.8612
mT5 (Base)	ViHSD	0.8746	0.8877	0.6246
mT5 (Base)	ViCTSD	0.8932	0.9024	0.7053
mT5 (Base)	ViHOS	0.9075	0.8957	0.8501
ViHateT5 (Ours)	ViHSD	**0.8815**	**0.8849**	**0.6698**
ViHateT5 (Ours)	ViCTSD	**0.9105**	**0.9158**	**0.7189**
ViHateT5 (Ours)	ViHOS	**0.9081**	**0.9055**	**0.8616**

Table: T5 model comparison (Table 4 – Paper Reproduction)

Model	ViHSD	ViCTSD	ViHOS	Avg F1
ViHateT5 (Ours)	**0.6698**	**0.7189**	**0.8616**	**0.7501**
ViT5 (Base)	0.6625	0.7163	0.8612	0.7467
mT5 (Base)	0.6246	0.7053	0.8501	0.7267

Table: Average Macro F1 for T5 models

Pre-training Impact (Table 5)

100K Hate-Only Pre-training

Dataset	Acc	F1-W	F1-M
ViHSD	0.8789	0.8784	0.6808
ViCTSD	0.9070	0.9283	0.6586
ViHOS	0.9039	0.8981	0.8541

200K Balanced Pre-training

Dataset	Acc	F1-W	F1-M
ViHSD	**0.8815**	**0.8849**	0.6698
ViCTSD	**0.9105**	0.9158	**0.7189**
ViHOS	**0.9081**	**0.9055**	**0.8616**

Pre-training Setup	ViHSD	ViCTSD	ViHOS	Avg F1
200K Balanced	0.6698	**0.7189**	**0.8616**	**0.7501**
100K Hate-Only	**0.6808**	0.6586	0.8541	0.7312

Table: Pre-training impact comparison (Table 5)

⇒ **200K balanced** pre-training yields better overall performance (+1.9% avg F1).

Extended BERT Comparison (Table 6)

Multilingual Models

Model	Acc	F1-W	F1-M
xlm-roberta-large	0.9204	0.7755	0.7968
xlm-roberta-base	0.9189	0.7722	0.8028
distilbert-multi	0.9115	0.7459	0.7754
bert-multi-uncased	0.9102	0.7557	0.7784
bert-multi-cased	0.9094	0.7548	0.7740

Monolingual (Vietnamese) Models

Model	Acc	F1-W	F1-M
visobert	**0.9296**	**0.8051**	**0.8128**
phobert-large	0.9245	0.7895	0.7832
phobert-base-v2	0.9216	0.7888	0.7810
vibert-base-cased	0.9117	0.7385	0.7771
phobert-base	0.9231	0.7562	0.7764

Key Finding

ViSoBERT achieves the highest Macro F1 (**0.8128**) among all 10 BERT variants.
Monolingual pre-training on Vietnamese social media data provides a clear advantage.

Results Summary

Key Findings

1. **ViHateT5 (Ours)** achieves the best T5 avg Macro F1 = **0.7501**, outperforming ViT5 (0.7467) and mT5 (0.7267)
2. **200K balanced pre-training** outperforms 100K hate-only (+1.9% avg F1)
3. **ViSoBERT** is the best encoder model (avg F1 = 0.7498, extended F1 = 0.8128)
4. Auto-labeling achieves **97.5% accuracy** on 10.7M samples

Best Models Per Task

- **ViHSD**: ViHateT5 (F1 = 0.6698)
- **ViCTSD**: ViHateT5 (F1 = 0.7189)
- **ViHOS**: ViHateT5 (F1 = 0.8616)

Observation

- T5 text-to-text approach slightly edges BERT baselines
- Unified model vs. 8 separate BERT classifiers
- Domain pre-training is more impactful than model size

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Limitations & Future Work

Current Limitations

- Improvement is purely **data-centric**; model architecture unchanged
- Minority classes (e.g., OFFENSIVE in ViHSD) remain challenging
- Auto-labeling may **propagate biases** from ViSoBERT into pre-training data
- Limited to Vietnamese language only

Future Directions

- **Model-centric improvements**: focal loss, curriculum learning, architecture modifications
- Deeper **error analysis** and failure case studies
- **Data augmentation** for underrepresented classes
- Larger pre-training corpus and longer schedules
- Cross-lingual transfer for other low-resource languages

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Summary

- Successfully **reimplemented ViHateT5** (ACL 2024) with reproducible results
- Built an **auto-labeling pipeline** (ViSoBERT, 97.5% accuracy, 10.7M samples)
- Demonstrated the value of **data-centric improvement**: balanced pre-training yields +1.9% avg F1 over hate-only
- Comprehensive benchmarking: **8 BERT** + **3 T5** models across **3 datasets**
- All models and datasets published on **HuggingFace**

Resources

- **GitHub**: `https://github.com/paht2005/DS200.Q21-Group2-Reimplementation-and-Improvement-of-ViHateT5-for-Vietnamese-HSD-Project`
- **HuggingFace**: `https://huggingface.co/collections/NCPhat2005/ds200q21-big-data-analysis-group-2`

Thank you! Questions?

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